

Yashvardhan Singh

signinwithyash1178@gmail.com | +91 7976780495

[LinkedIn](#) | [X.com](#) | [Github](#) | [LeetCode](#) | [Codeforces](#) | [Portfolio](#)

SKILLS

Languages: C/C++, Python, JavaScript, TypeScript

Cloud & DevOps: AWS (EKS, ECS Fargate, ECR, VPC, ALB, EC2, IAM), Docker, Kubernetes, CI/CD (GitHub Actions), Bash Scripting

Web & App Technologies: Next.js, React.js, Node.js, NestJS, Express.js, MongoDB, SQL, REST APIs, WebSockets, Socket.IO, React Query, TanStack (Query, Router, Table), Firebase, ElectronJS, noVNC, WebRTC, OAuth 2.0

Other Skills: DSA, OOP, DBMS, Computer Networks, Operating Systems, System Design, Microservices Architecture, SOLID & DRY Principles, API Design & Authentication, AI/ML

WORK EXPERIENCE

Cloud/DevOps Intern | *e-Yantra, IIT Bombay* May 2026 - Jul 2026

- Designed and built a **Kubernetes-based multi-tenant** architecture provisioning isolated, on-demand compute environments.
- Implemented **WebRTC-based remote** desktop streaming for GUI-heavy applications; set up a **full observability** stack (Prometheus, Grafana, Loki).
- Hardened the platform (HttpOnly cookies, **CSRF protection**, rate limiting, **RBAC**) and load-tested for concurrent multi-user scenarios.

Backend Engineer (Collaborator) | *Dezai* 2026

- Contributed as a backend collaborator on a full-stack platform, **shipping 7+ merged** pull requests across auth, caching, and admin systems.
- Caching & Resilience** — Built a fail-fast Redis client with circuit breaker and **stampede protection**; replaced deprecated caching library with a Redis-backed cache-aside pattern.
- Auth & Security** — Added role guards to 17 previously public API endpoints; fixed login redirect flow; pinned **NextAuth** version to avoid breaking updates.
- Data Integrity** — Fixed **missing DB migrations** for status enums.

R&D Intern | *NVIDIA - Jetson Nano & Orin* Jul 2025 - Aug 2025

- AI & robotics intern.

Member | *A.T.O.M (Robotics & AI Society of MAIT)* Oct 2024 - Present

SOFTWARE & PROJECTS

AWS Cloud-Native Web Deployment (2026) | *AWS ECS Fargate, Docker, VPC, ALB*

[GitHub](#)

- Engineered a production-grade, serverless deployment architecture using **AWS ECS Fargate**, configured with a custom **VPC**, strict **IAM** roles, and an **Application Load Balancer** (ALB) for secure traffic routing.
- Optimized containerization utilizing multi-stage Docker builds and cross-platform compilation (**buildx**), drastically reducing final image size and resolving local-to-cloud hardware architecture mismatches.

SCNARE v3: ROS 2 EKS Lab Platform (2026) | *NestJS, TypeScript, AWS EKS, Kubernetes API*

[GitHub](#)

- Developed a NestJS backend service to dynamically provision isolated, on-demand ROS 2 learning environments (namespaces/pods) within an **AWS EKS cluster**.
- Engineered a secure reverse proxy utilizing **RS256 JWTs** and an **OAuth2 sidecar** to authenticate and stream in-browser **ttty terminals** via an AWS ALB Ingress.

Work Order Processing System (2026) | *FastAPI, Celery, PostgreSQL/pgvector, Next.js, Redis*

[GitHub](#)

- Architected a multi-agent pipeline (**Orchestrator, RAG Retriever, Worker, Verifier**) that ingests work orders, retrieves context via hybrid **pgvector + full-text** search, and streams real-time decision traces to the frontend over **SSE**.
- Implemented cost-aware LLM routing with a **Redis-cached**, two-tier model system (cheap/fast vs. capable) enforcing a per-order **cost ceiling**, with every agent step logged for full observability.

EDUCATION

Maharaja Agrasen Institute of Technology (MAIT) | Bachelor of Technology in Information Technology 2024 - 2028

CGPA: 8.45/10

St. Paul's Sr. Sec. School, Ajmer 2020 - 2023

- Senior Secondary (XII-CBSE): 82.8%
- High School (X-CBSE): 91.8%

ACHIEVEMENTS

IICPC CodeFest (Rank 4085), ICPC 2025 (Kanpur Region) (Rank 2027), E-yantra-25 (IITB), HWM2.0 (2nd place)